SUMMER SCHOOL_

< PART 0 - SETUP AND TOOLING />

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> ”Tell me and I forget, teach me and I may remember, involve me and I learn.”
> — **Benjamin Franklin**

Program Overview & Objectives

Welcome to the Epitech Summer School. This document serves as your definitive environment blueprint. Every participant from absolute beginners to seasoned developers must initialize their workspace from this identical, reproducible foundation to ensure seamless execution.

To build the ”Automated Newsroom”, you will need specific software on your machine and a few free cloud accounts.

For Windows Users: We highly recommend installing the Windows Subsystem for Linux (WSL) to emulate a Linux environment. This will simplify your development experience and reduce potential compatibility issues.

- ✓ Open PowerShell as Administrator and run: `wsl --install`.
- ✓ Restart your computer. Upon restarting, a Ubuntu terminal should open automatically.
- ✓ **Account Setup:** The terminal will prompt you to create a default UNIX user account. Enter a username and a password. *(Note: When typing your password, nothing will show on the screen. This is a normal Linux security feature. Just type it and press Enter).*
- ✓ Once your account is ready, run the following commands to install Node.js (you will be asked to type the password you just created):

```
sudo apt update && sudo apt upgrade
```

Note: If you don't want to use WSL, you can just download the Windows installer directly from the Node.js website.

For macOS Users: You may need to install Xcode Command Line Tools. Open your terminal and run: `xcode-select --install`.

- ✓ Open Visual Studio Code from your Applications folder.
- ✓ Open the Command Palette by pressing Cmd + Shift + P.
- ✓ Type shell command in the search bar that appears at the top.
- ✓ Select the option that says: Shell Command: Install 'code' command in PATH.
- ✓ You may be prompted to enter your Mac administrator password to allow the change.
- ✓ A small success notification will appear in the bottom right corner of the editor.

Now you can open VS Code from the terminal by typing `code .` in any directory.

Step 1: Desktop Software Installation

Before writing a single line of code, you must equip your local operating system with the essential tools of modern web engineering. Follow the instructions below based on your computer's Operating System.

1. Visual Studio Code (The Code Editor)

We recommend a lightweight, extensible workspace capable of managing complex codebases.
- **Windows / macOS / Linux:** Go to Visual Studio Code to download and install the runtime.
- *Recommended Extensions:* Once installed, open the Extensions marketplace (`Ctrl+Shift+X` or `Cmd+Shift+X`) and add **Prettier - Code formatter** and **ESLint**.

2. Node.js & JavaScript Runtime

Modern web frameworks and server stacks rely on an asynchronous event-driven JavaScript runtime. Installing Node.js automatically equips your system with **npm** (Node Package Manager).

- ✓ **For macOS & Linux:**
  - Go to Node.js and download the **LTS (Long-Term Support)** version. Run the installer.
- ✓ **For Windows (Using WSL - Highly Recommended):**
  - Web development is much easier in a Linux environment. If you haven't already, follow the instructions above to install WSL and Ubuntu.
  - Open your new Ubuntu terminal and run the following commands to install Node.js:

    ```
    sudo apt update && sudo apt upgrade
    sudo apt install nodejs npm
    ```

  - *Note:* If you don't want to use WSL, you can just download the Windows installer directly from the Node.js website.

3. Postman (The API Testing Client)

When developing backend services, you need an independent workspace to craft HTTP requests.
- **Windows / macOS / Linux:** Go to Postman to download and install the graphical API platform interface.

Step 2: Cloud Ecosystem Accounts

Our application architecture relies on distributed cloud services for AI and automation. Create free-tier developer accounts on the following platforms:

1. The Automation Engine

You will use this to build the Event-Driven data pipelines.
- Go to Make.com and create a free account (Google Sign-In is recommended).

2. The AI Brain

You will use this platform to configure your Custom AI Assistant.
- Go to Google AI Studio and sign in with any free Google account.

3. The Local Tunnel

You will use this tool to expose your local code to the internet.
- Go to Ngrok and create a free account. Keep your login details safe, as you will need to copy your "Authtoken" later in the week.

Step 3: Local Environment Verification

Once installations are completed, open your operating system's native terminal (or the integrated terminal inside VS Code via `Ctrl+``) and execute the following boundary checks to ensure all tools are globally accessible:

```
# Verify the code editor CLI connection
code --version

# Verify the JavaScript runtime engine
node --version

# Verify the Node Package Manager engine
npm --version
```

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